Constraint Acquisition Motivated Model Counting Instances

Boris Wiegand
SHS – Stahl-Holding-Saar GmbH & Co. KGaA
boris.wiegand@stahl-holding-saar.de

March 25, 2024

Constraint programming, the holy grail of programming [1], separates the concerns of modeling a problem and finding a solution. As modeling the problem requires the rarely available combination of both domain knowledge and mathematical modeling expertise, acquiring constraints from data enables broader application of constraint programming [4]. Multiple constraint acquisition approaches [2, 6] propose using model counting to evaluate how well a constraint fits the data. As these methods need to evaluate many candidates during constraint search, model counting must be as fast as possible and should also work for larger problem sizes.

Therefore, we provide model counting instances of constraint programming puzzles, which are commonly used in constraint acquisition benchmarks. First, we give a 4×4 and a 9×9 Sudoku instance, where boolean variables assign numbers to cells and the given CNF instance express the constraints that each cell contains exactly one value and the values in each row, column and $n \times n$ block must be unique. While state-of-the-art model counters such as ApproxMC [5] and D4 [3] easily solve the 4×4 instance, we did not get a result within an hour for both methods on the 9×9 instance.

Next, we provide instances for the N-Queens problem with $N \in \{4, 5, 6, 7, 8\}$. Boolean variables assign queens to squares the CNF express the constraints that each queen is assigned to exactly one square and queens on the board a not allowed to attack each other.

Our last set of instances is based on the problem of scheduling N teams in a double round-robin competition. Boolean variables assign home team and away team to a match day. The CND constraints ensure that each team plays against each other and the first and second half of the matches must be symmetric. We give instances for $N \in \{4, \ldots, 8\}$.

References

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